

Desk controller

Table of contents

1	Root_sch
2	System_overview
3	Interface_Power_supply
4	Interface_servomotor
5	Interface_Limit_switches
6	Block_diagram
7	DCDC_converter_3V3
8	Microcontroller
9	Servomotor_interface
10	Limit_switch_interface
11	Display

Change history

1.0.0	23/09/2026	First release
1.1.0	18/10/2026	Migration to KiBot template
1.0.0	TBD	TBD
1.0.0	TBD	TBD
1.0.0	TBD	TBD
1.0.0	TBD	TBD
1.0.0	TBD	TBD
1.0.0	TBD	TBD
1.0.0	TBD	TBD

System overview



File: System_overview.kicad_sch



Desk controller

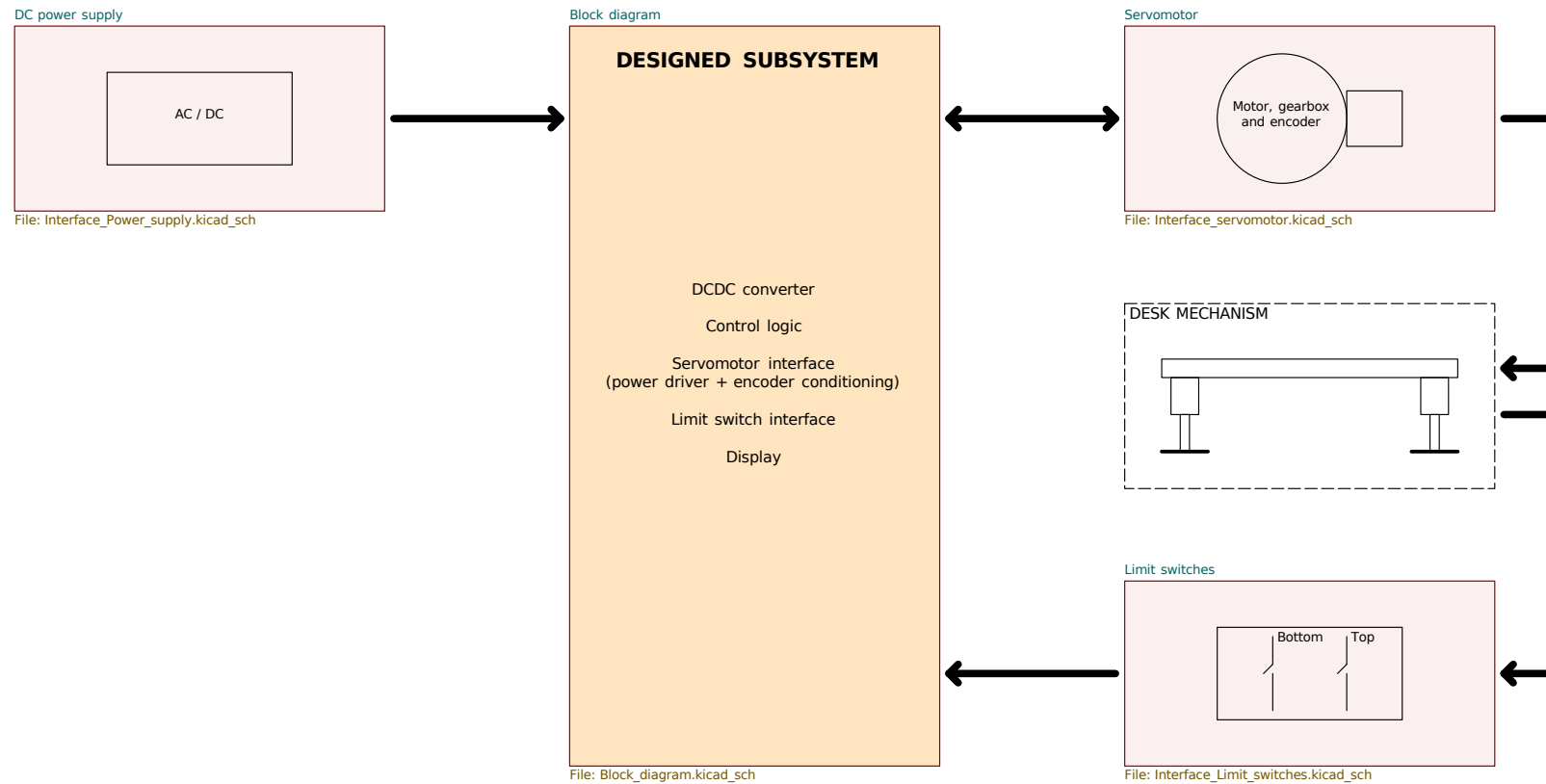
Notes

This is a comment 1 field - potentially not used
This is a comment 2 field - potentially not used
This is a comment 3 field - potentially not used
This is a comment 4 field - potentially not used

Sheet: Root
File: ktest.kicad_sch

Rev: 0.0.1	Drwn: FLR
Mat: Concept	Rev'd: FLR
Date: 2026-10-06	RLs'd: FLR
Dwg: NA	Size: A4
Generated with KiCad	Page: 1/11

System overview



This sheet shows the overall system and the interfaces to external components. External components are shown for system context only and are not part of this schematic. Only the "DESIGNED SUBSYSTEM" is covered by this schematic.

[Return to front sheet](#)



Desk controller

Notes

Sheet: System overview

File: System_overview.kicad_sch

Rev:

Mat: Concept

Date:

Dwg: NA

Generated with KiCad

Drwn: FLR

Rev'd: FLR

RLs'd: FLR

Size: A4

Page: 2/11

DC power supply

Interface specification

Barrel connector (Dell type, with ID pin)

Pin

- 1 Supply - 19.5 V, 3 A (inner shell)
- 2 ID (internal pin)
- 3 GND (outer shell)
- SH Shield

[Return to front sheet](#)



Desk controller

Notes

Sheet: DC power supply	
File: Interface_Power_supply.kicad_sch	
Rev:	Drwn: FLR
Mat: Concept	Rev'd: FLR
Date:	RLs'd: FLR
Dwg: NA	Size: A4
Generated with KiCad	Page: 3/11

Interface specification

JST B6B-XH-A / JST XHP-6 connector

Pin	Description
1	Encoder B output (VCC)
2	Encoder A output (VCC)
3	GND
4	VCC (3.3 - 5V)
5	Motor B (3 A, 12-24 V)
6	Motor A (3 A, 12-24 V)



Desk controller	Sheet: Servomotor	
	File: Interface_servomotor.kicad_sch	
	Rev:	Drwn: FLR
	Mat: Concept	Rev'd: FLR
	Date:	RLs'd: FLR
Notes	Dwg: NA	Size: A4
	Generated with KiCad	Page: 4/11

Limit switches

Interface specification

JST B4B XH A / JST XHP-4 connector

Pin Description

- 1 Bottom limit switch (Normally open, switch closes to GND)
- 2 GND
- 3 GND
- 4 Top limit switch (Normally open, switch closes to GND)

[Return to front sheet](#)

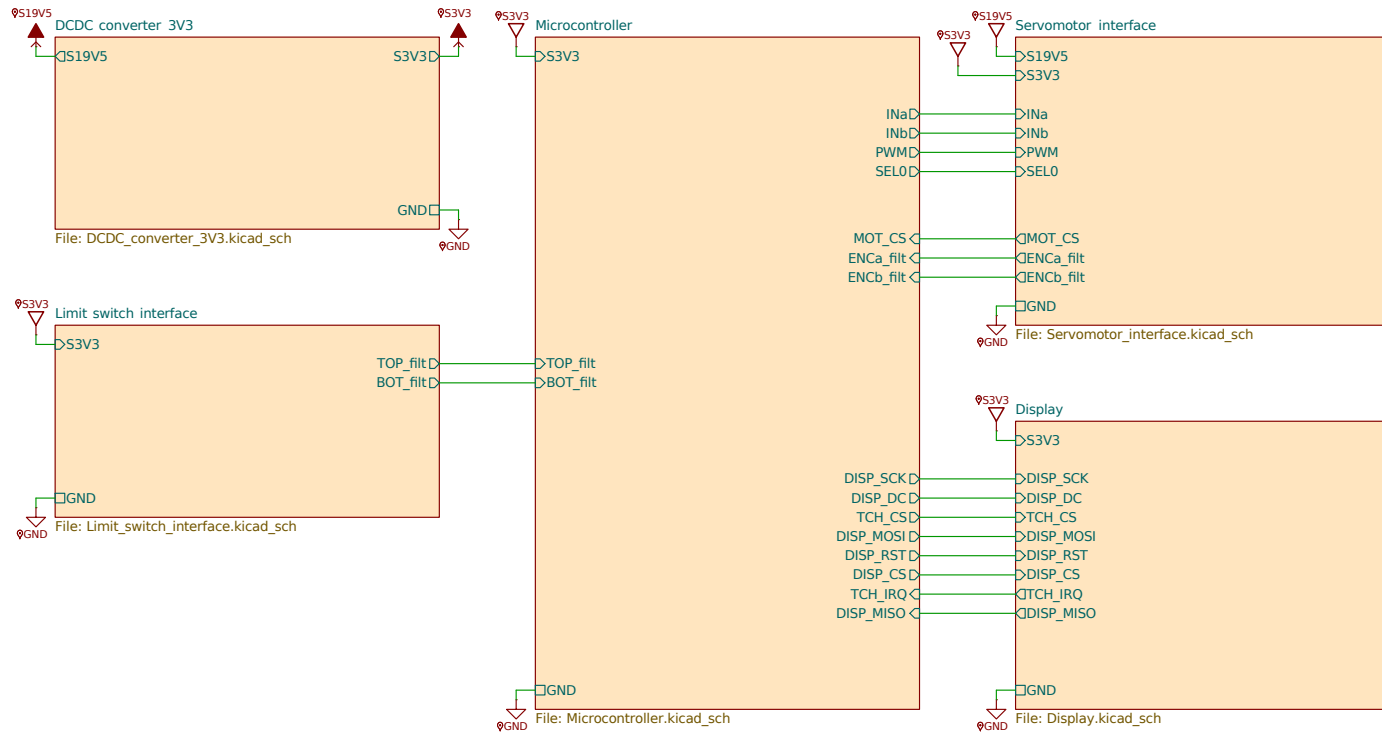


Desk controller

Notes

Sheet: Limit switches	
File: Interface_Limit_switches.kicad_sch	
Rev:	Drwn: FLR
Mat: Concept	Rev'd: FLR
Date:	RLs'd: FLR
Dwg: NA	Size: A4
Generated with KiCad	Page: 5/11

Block diagram



[Return to front sheet](#)



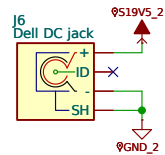
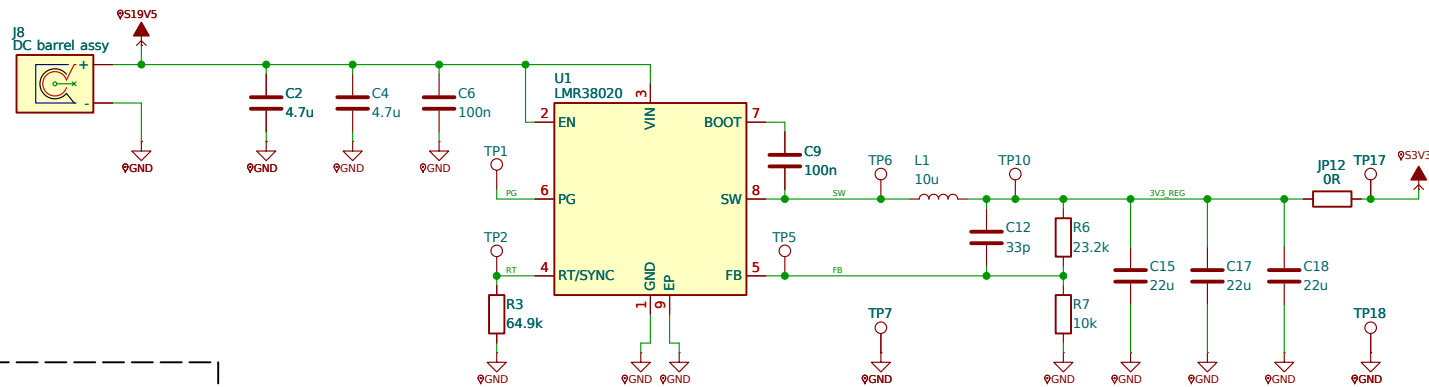
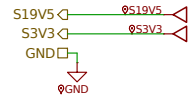
Desk controller

Notes

Sheet: Block diagram	
File: Block_diagram.kicad_sch	
Rev:	Drwn: FLR
Mat: Concept	Rev'd: FLR
Date:	RLs'd: FLR
Dwg: NA	Size: A4
Generated with KiCad	Page: 6/11

DCDC converter 3V3

Block interface



J6 is fitted to a small "standalone" PCB which will then be soldered into the "main" PCB. This standalone PCB + connector are what make the "DC barrel assy" used in J8 position.

[Return to front sheet](#)

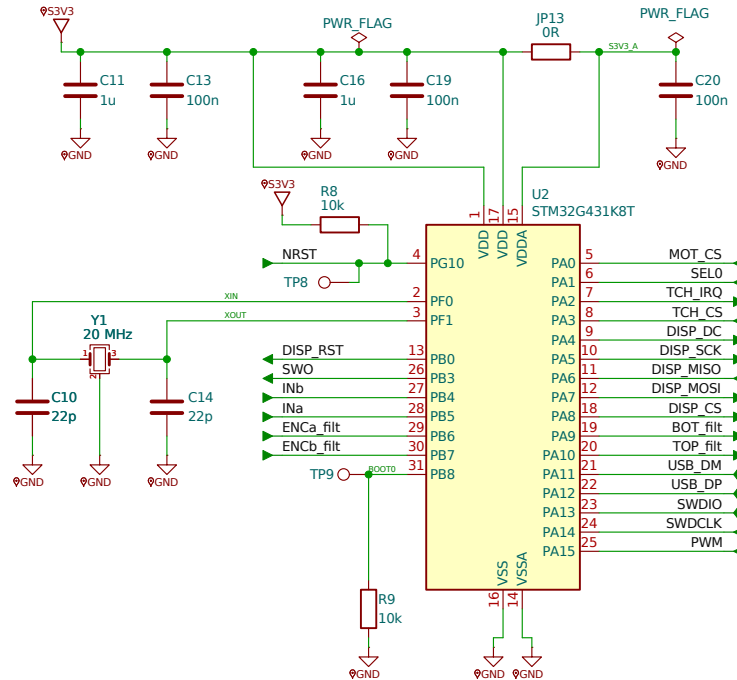
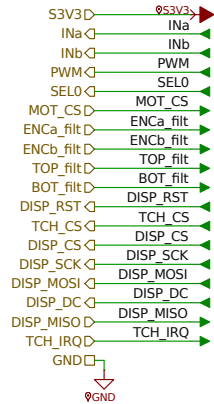


Desk controller

Notes

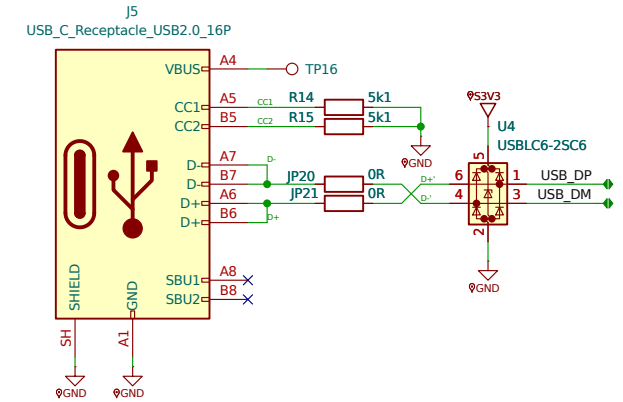
Sheet: DCDC converter 3V3	
File: DCDC_converter_3V3.kicad_sch	
Rev: Concept	Drwn: FLR
Date:	Rev'd: FLR
Dwg: NA	Rs'd: FLR
Generated with KiCad	Size: A4
	Page: 7/11

Block interface

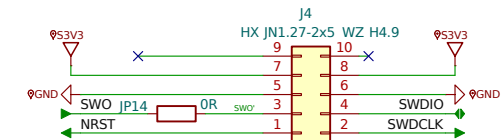


Microcontroller

USB interface (serial comms port)



SWD header



[Return to front sheet](#)



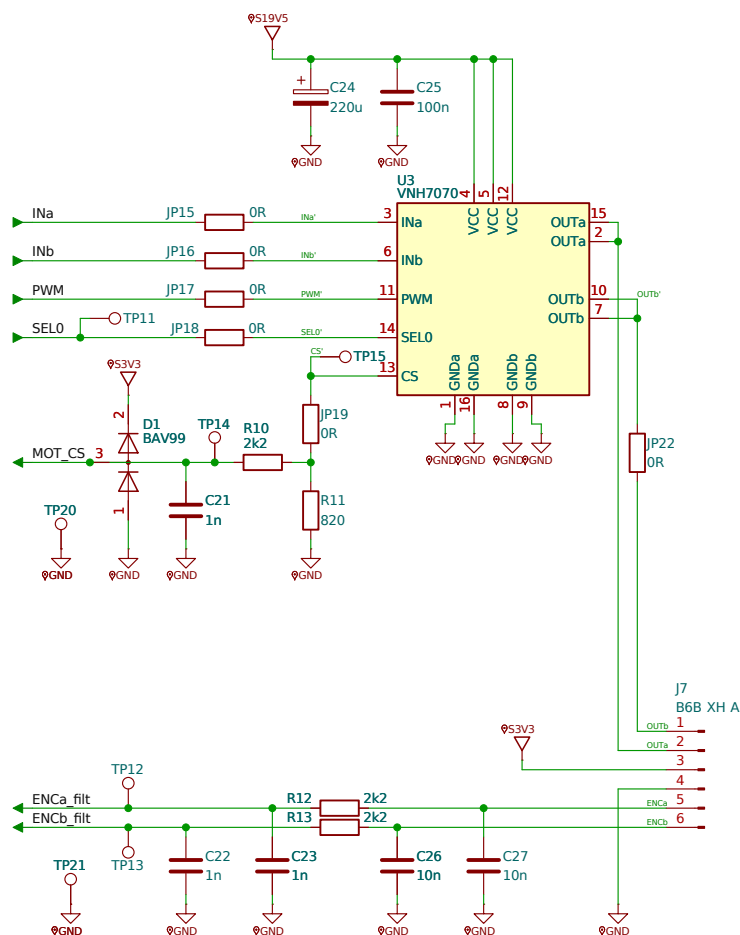
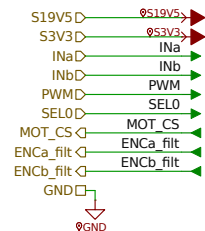
Desk controller

Notes

Sheet: Microcontroller	
File: Microcontroller.kicad_sch	
Rev:	Drwn: FLR
Mat: Concept	Rev'd: FLR
Date:	RLs'd: FLR
Dwg: NA	Size: A4
Generated with KiCad	
Page: 8/11	

Servomotor interface

Block interface



[Return to front sheet](#)



Desk controller

Notes

Sheet: Servomotor interface

File: Servomotor_interface.kicad_sch

Rev:

Mat: Concept

Date:

Dwg: NA

Generated with KiCad

Drwn: FLR

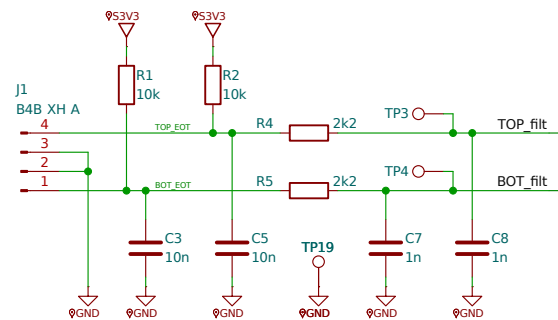
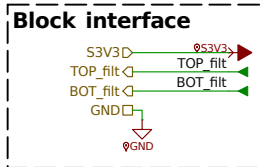
Rev'd: FLR

RLs'd: FLR

Size: A4

Page: 9/11

Limit switch interface



Return to front sheet



Desk controller

Notes

Sheet: Limit switch interface

File: Limit_switch_interface.kicad_sch

Rev:	Drwn: FLR
------	-----------

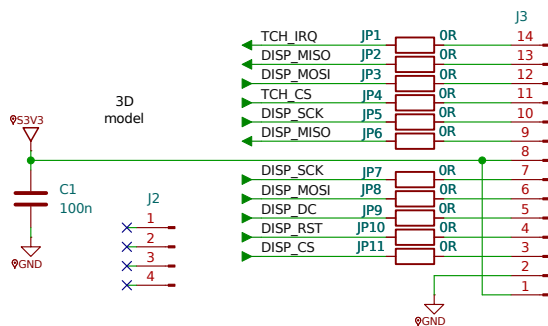
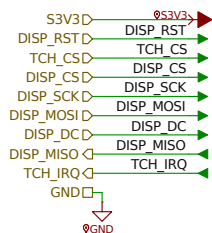
Mat: Concept	Rev'd: FLR
--------------	------------

Date: Rls'd: FLR

Dwg: NA	Size: A4
---------	----------

Generated with KiCad	Page: 10/11
----------------------	-------------

Block interface



TFT1
TFT_2.4in_320x240



The display will not be directly soldered to the PCB. Instead, generic 2.54 mm pitch sockets J2 and J3 will be used as connectors.

[Return to front sheet](#)



Desk controller

Notes

Sheet: Display
File: Display.kicad_sch

Rev: Concept	Drwn: FLR
Date:	Rev'd: FLR
Dwg: NA	RLs'd: FLR
Generated with KiCad	Size: A4
	Page: 11/11